# AEGIS-RF

## Confidence-Aware, Resource-Adaptive RF Spectrum Intelligence on PolarFire SoC

Technical Project Proposal — Microchip PolarFire® FPGA Design Contest 2026

**CONTEST SUBMISSION METADATA**
**Competition Track: Track 2 — PolarFire SoC Icicle Kit**
**Target platform/device: confirm exact kit revision and device identifier against the official kit documentation**
**Design tools and software: use versions supported by the selected kit/BSP; confirm before implementation**
**Document purpose: technical project proposal and architecture specification**
**Official proposal deadline in supplied rules: October 3, 2026, 11:59 PM PDT. Do not treat October 24 as official unless a newer organiser notice confirms an extension.**
**Development window in supplied rules: November 30, 2026 – March 27, 2027**
**Submission status: team to confirm. Do not alter a previously submitted package unless organisers permit it.**

Table 1. Project Team & Submission Administrative Information

| Item / Parameter | Proposal Specification / Status |
|---|---|
| Project Acronym & Title | AEGIS-RF: Confidence-Aware, Resource-Adaptive RF Spectrum Intelligence |
| **Target Platform** | PolarFire SoC Icicle Kit (exact board revision and orderable identifier to be confirmed) |
| Competition Track | Track 2 — Connected Real-Time Systems Using PolarFire SoC Icicle Kit |
| Primary Implementation Focus | Closed-Loop FPGA DSP + RISC-V Embedded Linux Supervisory Co-Design |
| Core Innovation | Confidence-Aware Downstream Workload Adaptation via Synchronized AXI-Lite Registers |
| Primary Verification Harness | Bit-Accurate Synthetic MATLAB Complex I/Q Test Vectors (Reproducible) |
| Secondary Hardware Extension | Optional Recorded ADALM-PLUTO (PlutoSDR) Over-the-Air Captures |
| Team Composition | Registered Multi-Disciplinary Engineering Team (Contest Track 2 Participants) |
| Team Lead Contact & Address | Verified in Official Contest Portal Registration Entry |
| Official Submission Drive Link | Refer to Submitted Public Google Drive / OneDrive Repository Link |

## 1. Executive Summary

Modern edge radio frequency (RF) spectrum monitoring operates under volatile and unpredictable conditions: fluctuating channel occupancy, intermittent burst transmissions, non-stationary noise floors, unknown signal-to-noise ratios (SNR), and agile interference. Conventional edge receivers rely on static digital signal processing (DSP) pipelines configured permanently for worst-case high resolution. This architectural rigidity incurs substantial, continuous computational overhead and energy dissipation during quiet spectral intervals, while fixed low-complexity architectures risk missing transient, weak, or narrow-band signals in congested bands.

AEGIS-RF addresses this fundamental limitation by establishing a heterogeneous, confidence-aware, resource-adaptive RF edge receiver on the Microchip PolarFire® SoC Icicle Kit (MPFS250T). The system partitions operations into two decoupled execution loops: a deterministic, streaming Real-Time FPGA DSP Path (the 'Fast Loop') and an intelligent RISC-V Linux Supervisory Control Path (the 'Slow Loop').

**OFFICIAL PORTAL PROJECT OVERVIEW (MAXIMUM 300 WORDS — RECOUNT BEFORE PORTAL ENTRY)**

Modern edge radio frequency (RF) spectrum monitoring operates under volatile conditions: fluctuating signal occupancy, dynamic interference, fading, and unknown signal-to-noise ratios. Conventional monitoring receivers deploy static signal processing pipelines configured permanently for worst-case high resolution. This incurs constant, unnecessary computational workload and power consumption during quiet spectral intervals, yet fixed low-complexity architectures miss transient or weak signals in dense environments.

AEGIS-RF (Confidence-Aware, Resource-Adaptive RF Spectrum Intelligence on PolarFire SoC) investigates the core engineering question: *Can a confidence-aware, resource-adaptive RF edge receiver reduce measured energy consumption or processing workload while maintaining defined signal-detection performance under changing interference conditions?*

Proposed for the PolarFire SoC Icicle Kit (MPFS250T), AEGIS-RF partitions operations across a deterministic hardware streaming path and a supervisory software path. The PolarFire FPGA fabric executes continuous, sample-rate streaming signal processing: I/Q sample buffering, a fixed 2048-point Microchip CoreFFT (subject to IP configuration verification), spectral power computation, continuous full-band candidate scanning, and windowed Cell-Averaging CFAR (CA-CFAR) detection.

The proposed DMA path will transfer detection reports and sparse feature vectors to shared LPDDR4 memory. Executing on the quad-core 64-bit SiFive U54 RISC-V Linux cluster, a Confidence-Aware Adaptive Controller evaluates detection uncertainty, signal persistence, and spectral occupancy. The controller dynamically scales downstream processing via AXI-Lite registers—modifying the extent of downstream candidate analysis (proposed 256/1024/2048-bin windows), detection threshold multipliers ($K_{CFAR}$), and reporting cadence at frame boundaries, while keeping the continuous 2048-bin full-band scan permanently active.

Primary validation employs reproducible MATLAB-generated complex I/Q test vectors (noise, multi-tone, BPSK, QPSK, frequency offsets, interference) with optional recorded SDR captures as an extension. Detection probability and false-alarm rate will be measured across modes alongside power changes on supported monitored rails via the onboard PAC1934, establishing an empirical foundation for cognitive edge RF receivers.

Table 2. AEGIS-RF Key Specifications & System Architecture Baseline

| System Parameter | Baseline Design Specification |
|---|---|
| **Target SoC Device** | PolarFire SoC MPFS250T-FCVG484E as named in the Icicle Kit User Guide; verify the exact fitted device suffix and board revision before build. |
| FPGA Logic Elements | 254K logic elements documented for the Icicle Kit device. Other fabric-resource capacities and utilisation remain TBD until the exact device datasheet and synthesis reports are checked. |
| RISC-V Subsystem (MSS) | Five-core MSS: one E51 monitor core plus four U54 application cores. Use a supported Linux/BSP on U54 for supervisory control; exact clocks/configuration are to be confirmed. |
| Operating System | Use the Linux/BSP and device-tree combination supported by the selected official reference-design release; do not pin an unverified kernel version. |
| On-Chip Memory | Board guide/brochure document 2 GB LPDDR4 ×32 on the MSS memory interface, plus 1 Gb SPI flash and 8 GB eMMC/SD-card option. Shared-buffer access from fabric is conditional on a verified AXI/DMA path. |
| FFT Core Architecture | Option B: fixed 2048-point CoreFFT, if supported by the selected IP configuration, with adaptive downstream workload |

| System Parameter | Baseline Design Specification |
|---|---|
| Adaptive Downstream Workload | Keep a low-cost candidate scan across all 2048 output bins active in every mode; adapt candidate-window detail, feature calculations and report cadence only. |
| Detection Engine | CA-CFAR with a calibrated threshold; keep it fixed for the primary comparison or calibrate configurations to a comparable Pfa |
| Primary Feature Set | Primary features: power spectrum, peak frequency/power, candidate bandwidth, detector margin, and persistence |
| Secondary Classification Features | Optional secondary features: envelope variance/PAPR or phase/IQ features, only if implemented and separately validated |
| Supervisory Intelligence | Confidence-aware controller using candidate quality, persistence, occupancy history, and hysteresis |
| Hardware Control Interface | Memory-mapped control registers only after bus connectivity is confirmed; apply staged settings at a validated frame boundary |
| Hardware Telemetry & Power | PAC1934 measures VDD, VDD25, VDDA25 and VDDA through MSS I2C. Do not label these as LPDDR4 or total-board power without independent evidence. |
| Primary Verification Input | Reproducible MATLAB-generated signed complex I/Q test vectors with documented scaling, sample rate, seeds, and ground truth |
| Secondary Hardware Extension | Optional recorded ADALM-PLUTO I/Q replay; not required for the minimum viable demonstration |

## 2. Problem Statement

Radio frequency environments are inherently non-stationary. In modern tactical, industrial, and commercial wireless deployments, the electromagnetic spectrum fluctuates dynamically across time, frequency, and space. Signals of interest (SOIs) are frequently accompanied by wideband thermal noise, unknown multipath fading, frequency offsets, agile frequency-hopping transmitters, and intentional or unintentional co-channel interference.

The Limitations of Fixed-Configuration Spectrum Monitoring: Current edge-deployed RF spectrum monitoring instruments exhibit a critical architectural compromise:

- Fixed high-detail processing can perform avoidable downstream work when the spectrum is quiet, but the magnitude of any excess workload or energy depends on the implementation and must be measured.
- A fixed low-workload configuration may provide insufficient analysis for weak or transient signals. AEGIS-RF will test this limitation with labelled scenarios rather than assume that every low-complexity monitor misses such signals.
- A static configuration does not explicitly adapt its analysis workload to uncertainty. AEGIS-RF tests whether multi-frame event evidence can guide downstream analysis while controlling false alarms and avoiding missed candidates.

> **★ CORE ENGINEERING RESEARCH QUESTION**
>
> **The AEGIS-RF project is formulated to answer a specific, measurable engineering question:**
>
> "Can a confidence-aware, resource-adaptive RF edge receiver reduce measured energy consumption or processing workload while maintaining defined signal-detection performance under changing interference conditions?"
>
> This question establishes the empirical boundary of our investigation. Every architectural choice, hardware register, control threshold, and test benchmark in this proposal is designed to provide verifiable quantitative evidence addressing this inquiry.

## 3. Motivation and Real-World Relevance

Edge RF sensing can support power-constrained environments such as remote monitoring sites, industrial communication installations, research laboratories, and selected mobile platforms. The practical value of adaptive processing depends on the signal environment, the required detection operating point, and the amount of downstream computation that can actually be reduced. AEGIS-RF will test these trade-offs using reproducible I/Q inputs before making claims about deployment-scale energy savings.

Why PolarFire SoC is a suitable platform: the Icicle Kit combines FPGA fabric with a RISC-V processing subsystem, enabling a hardware/software partition in which streaming DSP can be implemented in fabric and supervisory decisions can run on the processor. The exact board interfaces, software support, and measurement paths used by this project will be confirmed against the selected kit revision and supported design tools.

- FPGA signal processing: the proposed fabric pipeline will implement fixed-point spectral processing and candidate detection using IP and RTL supported by the installed Libero SoC flow. Any comparison with other FPGA technologies must be supported by a cited, like-for-like source; no platform-level power advantage is assumed in the project results.
- RISC-V supervisory processing: the PolarFire SoC includes a five-core RISC-V subsystem with application cores intended for operating-system workloads and a monitor core. The exact BSP, Linux version, core configuration, and available software interfaces will be taken from the supported kit release and verified during bring-up.
- Parameter-driven adaptation (without DPR): the baseline design uses runtime control registers to change validated downstream processing parameters, not the synthesized FFT length. Register writes will be staged and applied at a defined frame boundary. The transition latency and absence of data corruption are verification targets; no sub-microsecond or glitch-free guarantee is made before measurement.
- Power measurement plan: the Icicle Kit User Guide documents PAC1934 current sensing for four rails—VDD, VDD25, VDDA25, and VDDA—read through the MSS I2C interface. These are not automatically equivalent to total-board power, LPDDR4 power, or a separately isolated FPGA-fabric rail. AEGIS-RF will report only the documented channels actually exposed by the selected board revision and will characterize their update rate and measurement boundary. If a total-system energy claim is desired, an external calibrated input-power meter will be used as a separate measurement boundary.

## 4. Objectives and Measurable Success Criteria

AEGIS-RF defines measurable development objectives. All numerical performance values are targets to be tested, not achieved results. Timing closure, throughput, detection performance, energy, and resource utilisation will be reported only after the corresponding simulation, synthesis, or hardware measurements are completed.

Table 3. AEGIS-RF Project Objectives & Verifiable Acceptance Criteria

| Objective | Measurable Acceptance Criteria & Success Metric |
|---|---|

| Objective | Measurable Acceptance Criteria & Success Metric |
|---|---|
| 1. Bit-Accurate Verification Model | Create reproducible MATLAB I/Q test vectors and a floating-point reference. Select fixed-point word lengths, then report measured NMSE against a documented tolerance; do not guarantee < -50 dB before modelling. |
| 2. FPGA Streaming DSP Pipeline | Implement the fixed-length FFT, power calculation, full-band candidate scan and candidate-analysis path. Timing closure at the selected clock is a target to be verified in Libero reports. |
| 3. Decoupled Memory & Control Path | Implement a supported descriptor transfer and buffer-ownership protocol. Demonstrate sustained operation without sample loss under CPU and telemetry load. |
| 4. Confidence-Aware Controller | Implement the confidence controller and staged register updates. Verify escalation, de-escalation, reset, timeout, and safe fallback in simulation and hardware. |
| 5. 3-Way Comparative Evaluation | Measure Pd, Pfa, latency, throughput, resource use, CPU workload and energy for fixed high-detail, fixed lower-workload and adaptive configurations using comparable detector operating points. |

# 5. Proposed Solution

Architectural Overview: AEGIS-RF implements a closed-loop heterogeneous architecture partitioned into two distinct domains:

- Fast Loop (FPGA DSP path): accepts framed complex I/Q data, executes a fixed 2048-point FFT if the selected IP configuration supports it, computes spectral power, performs a full-band low-cost candidate scan, and generates compact candidate descriptors. More expensive CFAR refinement and feature extraction are applied to candidate regions according to the active mode. Exact pipeline stages and data widths will be finalised after IP configuration and fixed-point modelling.
- Slow Loop (RISC-V supervisory path): reads candidate descriptors through a verified memory/data-transfer path, evaluates event confidence and occupancy history, and requests validated downstream processing configurations through memory-mapped control registers. The design will use only interfaces and DMA mechanisms confirmed to be supported by the chosen kit and design flow.

The hardware pipeline is designed to continue processing independently of Linux scheduling, provided the input source, FIFO depth, memory path, and throughput budget are correctly engineered. This independence must be demonstrated with sustained-stream tests under CPU and telemetry load; it is not a guarantee against buffer underruns without verification.

# 6. Innovation and Differentiation

## 6.1 Confidence-Aware Adaptive Processing Controller

The proposed contribution is a confidence-aware supervisory controller that selects among validated downstream workloads using multi-frame evidence. FFT, power-spectrum calculation, and CFAR are established building blocks; the differentiating hypothesis is that calibrated event confidence can guide additional analysis while preserving full-band candidate awareness and meeting a defined detection operating point.

The controller operates according to a formal five-stage cognitive pipeline:

- 1. Feature Ingestion: Receives compact feature vectors (peak frequency, peak power, occupied bandwidth, noise estimate, spectral moments, envelope variance) from the FPGA via DMA.
- 2. Uncertainty evaluation: computes a bounded event-quality score from peak-to-noise ratio, temporal persistence, and detector margin. The score is a control feature, not a probability of correct detection unless calibration demonstrates probabilistic reliability.
- 3. Workload selection: combines candidate presence, event-quality score, occupancy, and recent state to select Coarse Surveillance, Standard Candidate Analysis, or Detailed Candidate Analysis.
- 4. Targeted escalation: a candidate with low or unstable confidence, weak detector margin, or rapidly changing features triggers more detailed analysis. High confidence alone does not trigger escalation; it may support de-escalation only when the event is stable and occupancy is low.
- 5. Audit logging: records mode, transition reason, candidate features, frame index, decision time, detection outcome, and measured power data only when the corresponding telemetry is available.

> **OFFICIAL PORTAL INNOVATION & EXPECTED IMPACT (MAXIMUM 250 WORDS — RECOUNT BEFORE PORTAL ENTRY)**
>
> **Innovation:**
> AEGIS-RF shifts spectrum monitoring from static pipelines to an uncertainty-driven, closed-loop heterogeneous architecture. Rather than relying on rigid fixed parameters or high-risk Dynamic Partial Reconfiguration (DPR), AEGIS-RF integrates standard blocks with an innovative Confidence-Aware Adaptive Processing Controller on embedded RISC-V Linux. The continuous 2048-bin full-band candidate scan remains active in all modes, while the controller dynamically scales the extent of downstream candidate analysis (proposed 256/1024/2048-bin windows), detection threshold multipliers ($K_{CFAR}$), and reporting cadence via AXI-Lite registers at frame boundaries. Workload escalates during ambiguous events and downscales during quiescent periods, while detection probability and false-alarm rates are explicitly tracked to preserve detection fidelity.
>
> **Potential Impact:**
> Edge RF receivers in tactical sensing, industrial IoT, and wireless infrastructure operate under strict energy limits. AEGIS-RF provides an empirical methodology to evaluate whether runtime adaptation reduces workload and monitored rail power compared to fixed baselines. Demonstrating this closed loop on the PolarFire SoC delivers a reproducible blueprint for cognitive edge receivers.
>
> **Key Differentiators:**
> 1. **Confidence-Aware Control:** Workload scales based on quantified multi-frame detection confidence rather than naive thresholding.
> 2. **Defensible Fixed FFT Architecture:** Fixed 2048-point CoreFFT (subject to IP verification) with adaptive downstream analysis, avoiding unverified runtime FFT-length switching claims.
> 3. **Decoupled Heterogeneous Execution:** The FPGA streaming path is designed to operate independently of Linux scheduling, subject to validated buffering, flow control, and memory constraints.
> 4. **Grounded Power Telemetry:** PAC1934 rail measurements evaluate power changes on supported monitored rails, replacing unverified total-system energy claims.

Table 4. System Differentiation: AEGIS-RF vs. Contemporary Architectures

| Architecture Category | Conventional Implementation | AEGIS-RF Proposed Co-Design |
|---|---|---|
| Standard FPGA Spectrum Monitors | Some systems use fixed FFT/CFAR settings; behaviour varies by product. Avoid generalising without a named, cited implementation. | Proposed confidence-guided downstream workload adaptation while preserving full-band low-cost candidate awareness; benefit must be measured. |
| Software-Defined Radios (SDR) | SDR architectures range from host-only processing to hardware-accelerated designs; no single power or bandwidth profile applies to all. | Transfer compact candidate descriptors where feasible; measure data reduction and CPU workload on the actual implementation. |

| Architecture Category | Conventional Implementation | AEGIS-RF Proposed Co-Design |
|---|---|---|
| Edge AI RF Accelerators | Edge ML costs vary with model, quantisation, hardware, and input rate. | Use a small explainable controller or rule-based FSM; code size, memory, accuracy, and latency remain to be measured. |
| Reconfigurable FPGA Pipelines | Some FPGA systems use partial reconfiguration; latency and complexity depend on the device, bitstream, and design. | The baseline avoids DPR and uses staged runtime parameter updates. Transition correctness and latency require verification. |

## 7. System Block Diagram

The AEGIS-RF system architecture establishes clean isolation between real-time data movement, hardware DSP execution, and supervisory embedded software control. Figure 1 illustrates the end-to-end dataflow and control interfaces.

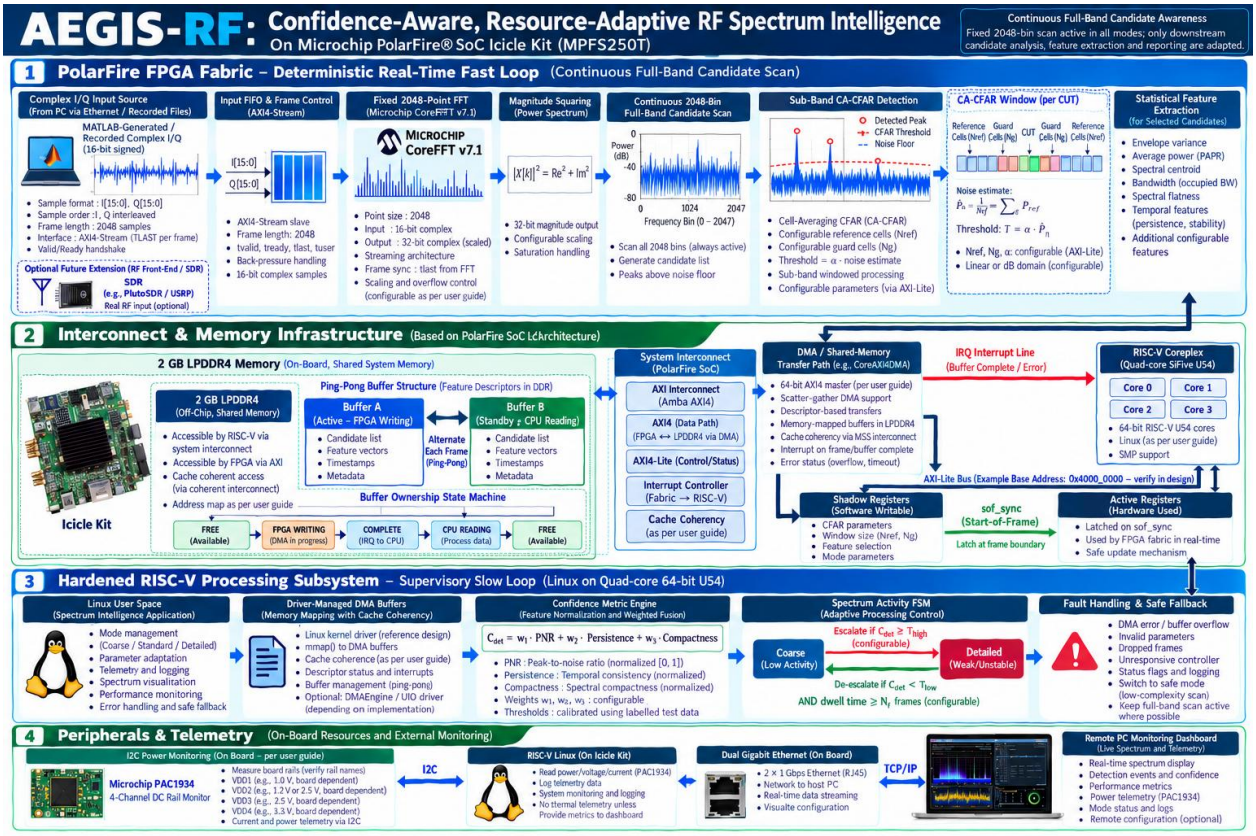



**Figure 1. AEGIS-RF Overall System Block Diagram Showing FPGA Fast Loop and RISC-V Slow Loop**

Architectural Components Breakdown: As depicted in Figure 1, the architecture incorporates twelve explicitly defined subsystems:

- 1. MATLAB I/Q synthesis: reproducible signed complex I/Q vectors with documented sample rate, scaling, scenario labels, random seeds, and ground truth.
- 2. Optional recorded RF input: previously recorded I/Q captures may be replayed if available and properly licensed; live PlutoSDR integration is an optional extension, not a dependency of the minimum viable demonstration.

- 3. Input buffering and alignment: a supported input/buffering path with framing and overflow detection. Use AXI4-Stream FIFO or clock-domain-crossing IP only after confirming the actual interface requirements.
- 4. FPGA Signal-Processing Pipeline: Pipelined streaming RTL blocks executing continuous sample transformation.
- 5. FFT and spectral power: a fixed 2048-point FFT configuration, subject to confirming the selected CoreFFT IP version, licence, interface, and supported data widths; followed by fixed-point power calculation.
- 6. CA-CFAR Detection Engine: Sliding window Cell-Averaging CFAR estimating local noise background and isolating candidate peaks.
- 7. Feature Extraction Unit: RTL statistical engine aggregating frequency-domain moments and time-domain envelope variance.
- 8. RISC-V confidence controller: supervisory software using the supported kit software stack; the final implementation language and Linux/BSP version will be selected during platform bring-up.
- 9. AXI-Lite Runtime Interface: Memory-mapped registers (0x00–0x1C) updating active hardware thresholds and bin selection.
- 10. Data transfer and shared memory: a verified DMA or supported transfer mechanism for compact descriptors. Do not assume a particular DMA controller or direct LPDDR4 path until the actual IP/interconnect design is confirmed.
- 11. Ethernet Event Logging: TCP/IP telemetry streaming detection reports and spectrum occupancy to a PC monitoring interface.
- 12. Power measurement: use the board's verified accessible monitoring channels; document the rail names, calibration, sample/update rate, and measurement boundary. Use an external meter if on-board telemetry cannot isolate the relevant power.

## 8. Detailed Hardware/Software Architecture

The PolarFire SoC is a suitable heterogeneous platform because the Icicle Kit combines an MPFS250T-FCVG484E-class device, documented as having 254K logic elements, with a five-core RISC-V processing subsystem, board LPDDR4, two Gigabit Ethernet ports, expansion interfaces, user LEDs/switches, and a four-channel PAC1934 current-monitoring device. AEGIS-RF assigns deterministic per-frame DSP to the FPGA fabric and supervisory adaptation, logging, and networking to software. It will reuse the vendor board constraints, MSS configuration, clocks, resets, and supported data-movement path wherever possible. The proposal does not assume that every documented board resource is available to the custom fabric design without checking bank ownership, muxing, voltage, timing, and the selected reference-design configuration. Claims about efficiency, timing, and resource headroom will be based on implementation reports and measurements rather than platform assumptions.

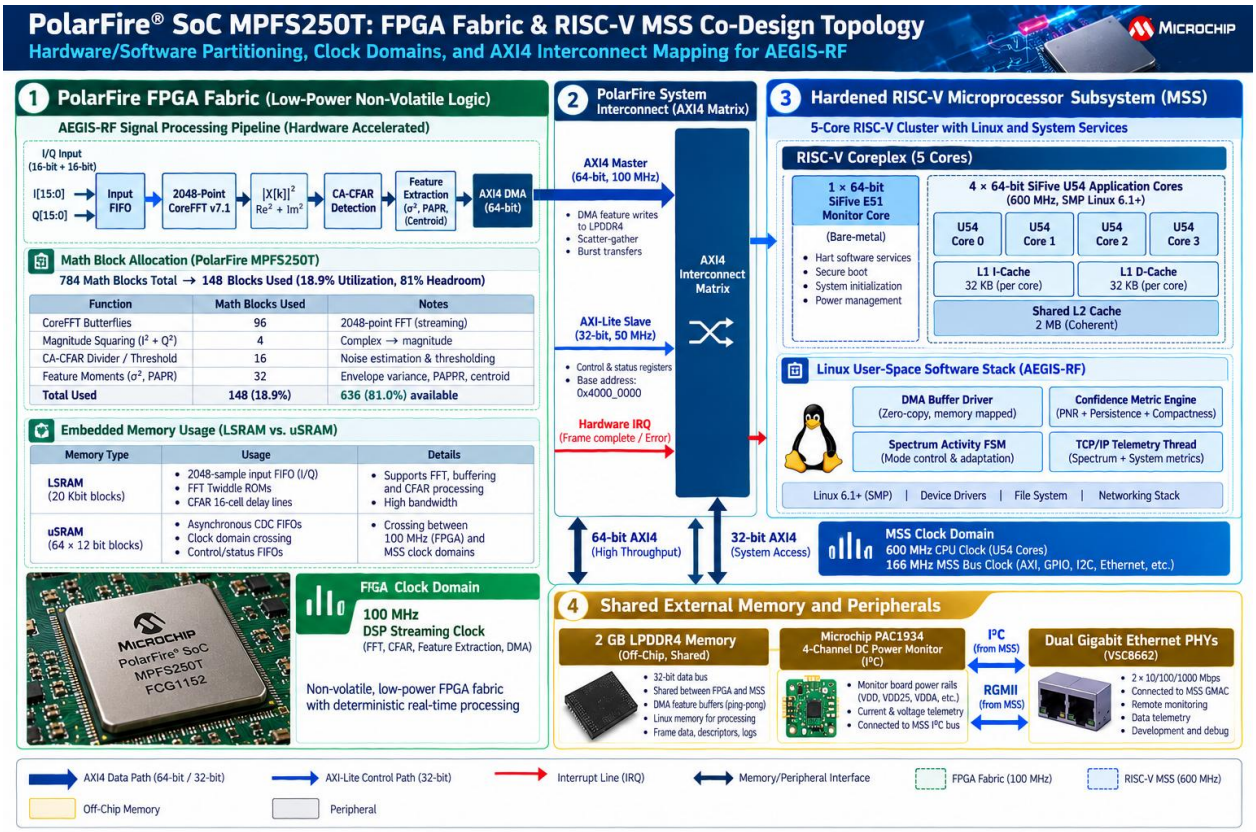



**Figure 2. FPGA–RISC-V Hardware/Software Partitioning and AXI Interconnect Topology**

Table 5. Hardware/Software Execution Domain Partitioning

| Subsystem / Block | Execution Domain | Implementation Rationale & Hardware Mapping |
|---|---|---|
| I/Q Input Buffer | FPGA Fabric (RTL) | Start with deterministic MATLAB-generated signed complex I/Q vectors replayed through a verified input path. The Icicle Kit has no native RF ADC; live RF capture is optional external hardware, not part of the MVP. |
| Microchip CoreFFT IP | FPGA Fabric (IP Core) | Fixed 2048-point FFT only after confirming the selected CoreFFT version, supported interface, data widths, scaling, and throughput. |
| Power Spectrum Unit | FPGA Fabric (RTL) | Read the documented PAC1934 channels VDD, VDD25, VDDA25 and VDDA over MSS I2C. These support rail-level comparisons only; use a calibrated external meter for total input power if required. |
| CA-CFAR Detection | FPGA Fabric (RTL) | CA-CFAR or another justified detector; define training/guard cells, edge handling, and calibrated threshold. |
| Feature Extractor | FPGA Fabric (RTL) | Extract candidate frequency, power, detector margin, bandwidth and selected statistical features; optional features may be gated by mode. |

| Subsystem / Block | Execution Domain | Implementation Rationale & Hardware Mapping |
|---|---|---|
| Descriptor Data Transfer | Fabric/bus interface (TBD) | Reuse the official AXI4_STREAM_DEMO/CoreAXI4DMAController pattern only after validating the selected release, interconnect, address map, interrupt path, Linux support and cache coherency. Use compact descriptors first. |
| LPDDR4 Memory | Board memory (access path TBD) | Memory allocation/access follows the selected kit/BSP. Verify physical addresses, cache coherency, buffer ownership and memory availability. |
| U54 RISC-V Cluster | Hardened CPU MSS | RISC-V application cores run the supported OS/controller configuration. Exact runtime and core allocation are implementation decisions. |
| E51 Monitor Core | Hardened CPU MSS | Monitor-core and HSS roles depend on the supported boot configuration; document the actual board/BSP arrangement. |
| AXI-Lite Controller | MSS AXI Master | AXI-Lite control path: add the AEGIS-RF register block to the generated SmartDesign and verify address decode, master/slave connectivity, clock/reset domains, access permissions and frame-boundary latching in simulation and on board. |
| PAC1934 Monitor | Onboard IC (I2C) | Read the documented PAC1934 channels VDD, VDD25, VDDA25 and VDDA over MSS I2C. These support rail-level comparisons only; use a calibrated external meter for total input power if required. |

## 8.1 Hardware Feasibility Anchored to the Official Icicle Kit Reference Design

AEGIS-RF will not start from a blank SoC design or assume that the reference bitstream already contains an RF pipeline. The Microchip polarfire-soc/icicle-kit-reference-design repository supplies a Libero SoC Tcl-generated baseline, board constraints, MSS configuration and a documented AXI4_STREAM_DEMO using CoreAXI4DMAController. The repository documentation identifies Libero SoC v2025.2+ as its current baseline; release notes should be checked before selecting a tag. The team will pin a compatible repository commit/tag, tool release, IP versions and the exact production/engineering-sample device variant. The board-specific five-document audit in Section 8.2 maps the documented resources to AEGIS-RF and identifies which capabilities are used by the minimum viable prototype versus deferred.

Feasibility strategy: first boot the unmodified supported reference design and verify serial/Linux access; next build and run the reference AXI4 streaming demo; then add AEGIS-RF blocks incrementally. Keep the first demonstrator independent of live RF hardware: replay deterministic MATLAB-generated I/Q vectors through a documented, bounded input path. Only after frame ordering, overflow detection, descriptor transfer and sustained throughput pass should the optional recorded SDR capture path be added. The exact CoreFFT version, transform size,

streaming handshake, fixed-point scaling, resource cost and licence must be confirmed in the installed Libero IP catalog before the 2048-point choice is frozen.

| Integration item | Official source / board fact | AEGIS-RF implementation action | Gate / evidence required |
|---|---|---|---|
| **Board / SoC baseline** | Icicle Kit User Guide identifies MPFS250T-FCVG484E, 254K logic elements, LPDDR4, Ethernet, four UARTs, SPI, SD/eMMC and expansion interfaces. | Confirm exact board orderable/revision and silicon variant; start from the matching official reference-design release. | Boot and peripheral smoke-test log; source tag, device and tool versions recorded. |
| **Libero / MSS baseline** | Official GitHub README describes Tcl-generated reference design for Linux and bare-metal configurations; current repository documents Libero SoC v2025.2+. | Generate from the official script; retain vendor-generated pin constraints, clocks, resets and MSS configuration unless a change is required and reviewed. | Successful generation, synthesis, place-and-route, timing report and board boot. |
| **Streaming / DMA path** | Repository includes AXI4_STREAM_DEMO using CoreAXI4DMAController; release notes document DMA-related fixes. | Treat this as a reusable example, not a guaranteed drop-in RF pipeline. Verify AXI4-Stream width/clock, DMA control/data path, interrupts, address map and Linux driver support. | Loopback/source-generator test first; counters show no overflow/underrun over a declared sustained duration. |
| **FFT / CFAR fabric** | Board provides FPGA fabric; the reference design does not itself establish AEGIS-RF CoreFFT or CA-CFAR availability. | Check IP catalogue/version/licence and simulate FFT handshake, scaling and latency. Implement candidate scan and CFAR in staged RTL/IP integration. | Bit-accurate MATLAB comparison, resource report, timing closure at selected clock and edge-bin/multi-candidate tests. |
| **Control / software** | MSS supports RISC-V software; reference design provides generated MSS and memory-map context. | Add a small AXI-Lite register slave and use a supported Linux/driver path; apply shadow registers only at a proven frame boundary. | Register read/write test, reset/interrupt tests, frame-boundary update tests, and no mid-frame configuration change. |
| **Memory / buffering** | Board guide documents LPDDR4 and its MSS interface; direct fabric-to-LPDDR4 sharing is not assumed. | Prefer descriptor transfer through the verified reference DMA/data-mover route. Document buffer ownership, address translation, cache policy and error recovery. | Address-map proof, cache-coherency test, bounded buffer design and sustained zero-loss measurement. |

| Integration item | Official source / board fact | AEGIS-RF implementation action | Gate / evidence required |
|---|---|---|---|
| **Power / energy** | User Guide documents PAC1934 measurement of VDD, VDD25, VDDA25 and VDDA over MSS I2C. | Measure only documented channels; do not call them LPDDR4 or total-board power. Pair with external calibrated input-power measurement if the desired boundary is not observable. | Record rail mapping, update rate, calibration, workload synchronisation and uncertainty. |

## 8.2 Board-Document Audit and Resource-to-Criteria Mapping

The following mapping is based on the five supplied board documents. It distinguishes board features that are documented from proposed AEGIS-RF uses and from integration assumptions that still require bench evidence. The board's high-speed PCIe lanes and expansion headers are not forced into the minimum prototype simply to claim utilisation; using a peripheral only when it improves the evaluation is stronger than adding unnecessary complexity.

| Board resource documented in supplied files | AEGIS-RF use and criterion served | Implementation boundary / verification gate |
|---|---|---|
| MPFS250T-FCVG484E-class PolarFire SoC; 254K logic elements (QuickStart/User Guide) | Implement the streaming DSP datapath in fabric: fixed FFT, spectral power, all-bin candidate scan, CFAR and selected features. Directly supports technical feasibility and FPGA relevance. | Do not invent LE/math-block/LSRAM totals from the kit guide. Obtain exact device datasheet and Libero reports; record LE, math-block, LSRAM, clock and timing results after synthesis/place-and-route. |
| Five-core RISC-V MSS: one E51 plus four U54 application cores (brochure/User Guide) | Run supervisory confidence FSM, occupancy history, hysteresis, event logging and dashboard networking on supported Linux/U54. Keeps the frame-rate control loop out of the sample-rate path. | Boot the matching reference image first; confirm BSP, kernel/device tree, drivers, user-space API and MSS/fabric register map. E51 is not assumed to run the Linux controller. |
| 2 GB LPDDR4 ×32 on MSS memory interface; 1 Gb SPI flash; 8 GB eMMC / SD option (brochure/User Guide/schematics) | Use LPDDR4 for Linux and bounded descriptor/frame buffers only through the verified data-movement route; use eMMC/SD for I/Q datasets, logs and reproducible test vectors. | Do not assume direct FPGA fabric access to LPDDR4. Prove AXI interconnect, DMA addressing, cache coherency, buffer ownership, alignment and overflow recovery. Start with compact feature descriptors. |
| Official reference design includes AXI4_STREAM_DEMO and CoreAXI4DMAController (GitHub reference design) | Use the vendor streaming/DMA example as the starting point for fabric-to-software data movement, reducing integration risk and improving reproducibility. | It is a reference example, not a prebuilt AEGIS-RF pipeline. Pin commit/tag and Libero release; first run the unmodified demo, then replace/extend its data source and sink in small steps. |

| Board resource documented in supplied files | AEGIS-RF use and criterion served | Implementation boundary / verification gate |
|---|---|---|
| 50 MHz fabric oscillator (User Guide §3.13; schematics) | Provide a known board clock reference for the fabric design; a derived DSP clock can support the proposed pipelined datapath if timing closure permits. | Do not claim a 100 MHz or any other target clock until CCC configuration and timing reports verify the design. Preserve vendor clock/reset constraints. |
| Two Gigabit Ethernet ports via VSC8662 (brochure/User Guide/schematics) | Use a supported Linux network interface for PC dashboard telemetry, experiment control and optional transfer of recorded I/Q files. Supports practical demo and impact. | Ethernet carries commands/results and file replay; it is not a raw RF front end. Benchmark telemetry enabled/disabled so it does not distort core DSP metrics. |
| PAC1934 current sensing on VDD, VDD25, VDDA25 and VDDA through MSS I2C (User Guide §3.7) | Log board-supported rail telemetry during A/B/C runs; correlate samples with mode and workload to test whether adaptive processing changes measured rail power. | These channels are not automatically total board, LPDDR4, or isolated fabric power. Verify the mapping, update rate, calibration and accessible driver. Use an external calibrated input-power meter for whole-board energy. |
| Four user LEDs and four push-buttons on HSIO Bank 0; USB-to-quad-UART bridge (User Guide/QuickStart) | Show current mode, candidate alarm, escalation and error state with LEDs/buttons; use serial ports for boot logs, debug and reproducible bring-up evidence. | Check the selected reference design's pin constraints and muxing before assigning them. Keep the UART console as the primary recovery/debug path. |
| 40-pin Raspberry Pi-compatible header and 16-pin mikroBUS (User Guide/brochure/schematics) | Reserve as optional expansion for a future external trigger, sensor, or supported peripheral if a real need arises. | Not needed for the minimum build; check bank voltages, pin assignment, shared functions and jumper configuration before attaching any module. No expansion hardware is assumed included. |
| PCIe Gen 2 root-port connector mapped to four SERDES lanes; one XCVR block/four lanes (User Guide/brochure) | Optional future extension for a suitable PCIe endpoint or higher-throughput peripheral, if it materially benefits the project. | Not part of MVP: adds endpoint, link training, driver and bandwidth-validation risk. Do not claim PCIe acceleration or usage in the baseline demo. |
| No RF ADC or SDR receiver listed among standard kit contents (QuickStart/brochure/User Guide) | Use MATLAB-generated, labelled signed complex I/Q vectors as the reproducible primary input; optional recorded PlutoSDR captures can extend realism later. | Live RF capture requires separately supplied external SDR/ADC and a verified host-to-board input path. It is not assumed available from the Icicle Kit itself. |

Board bring-up sequence: (1) inspect exact kit/device revision and jumper configuration; (2) boot the unmodified vendor image using the four UART ports; (3) build and run the matching official reference design; (4) pass its AXI4 stream/DMA demonstration; (5) add an I/Q frame source and FIFO/overflow counters; (6) integrate fixed FFT and compare against MATLAB; (7) add all-bin candidate scan and CFAR; (8) add compact descriptors and the supervisory controller; (9) measure power with a documented boundary; (10) run the held-out three-way benchmark. Every stage has a rollback point to the last known-good reference build.

# 9. Signal-Processing Methodology & FFT Architecture

FFT implementation: the baseline uses one fixed 2048-point transform, provided the selected IP configuration and tool release support the required interface and throughput. The transform length is treated as a synthesis-time design choice. Runtime adaptation changes only validated downstream work; it does not claim to convert the fixed core into a 256-, 512-, or 1024-point FFT.

Table 6. Evaluation of FFT Architecture Implementation Options

| Option | Architecture Description | Technical Assessment & Feasibility |
|---|---|---|
| Option A: Genuine Variable-Length FFT | Runtime switching among 256/512/1024/2048-point transforms within one core. | Do not use unless the selected IP explicitly supports it and the feature is verified; otherwise treat as unsupported/high risk. |
| Option B: Fixed 2048 FFT + Adaptive Downstream Workload (Selected) | One fixed 2048-point transform; full-band low-cost candidate scan remains active while detailed downstream analysis varies by candidate and mode. | Preferred baseline if the IP configuration supports the required interface. Energy/resource benefit must be measured; FFT cost remains in every mode. |
| Option C: Multiple Validated Configurations | Separate synthesised FFT configurations if tool/resource constraints permit. | Potentially higher resource and integration cost; assess with synthesis rather than assume it exceeds the budget. |

**FFT ARCHITECTURE AND FULL-BAND AWARENESS**

**1. Use a fixed 2048-point FFT only after confirming the selected IP/version and its interface, throughput, scaling, and licensing.**

**2. Keep a low-cost scan over all 2048 output bins active in every mode to identify candidate regions. Detailed feature extraction and optional refined processing may be limited to candidate windows in lower-workload modes.**

**3. The 256/512/1024 values refer to a downstream analysis span or candidate-window budget, not to the FFT transform length. Define how candidate windows are selected and how multiple or widely separated candidates are handled.**

**4. Do not claim energy savings from bin selection alone. Measure the actual power/workload difference; if the fixed FFT dominates energy, report that limitation.**

**5. Fallback: if controller or descriptor transfer fails, continue the validated full-band low-cost scan and flag degraded mode. Do not assume full-detail processing can be restored unless buffer and timing budgets support it.**

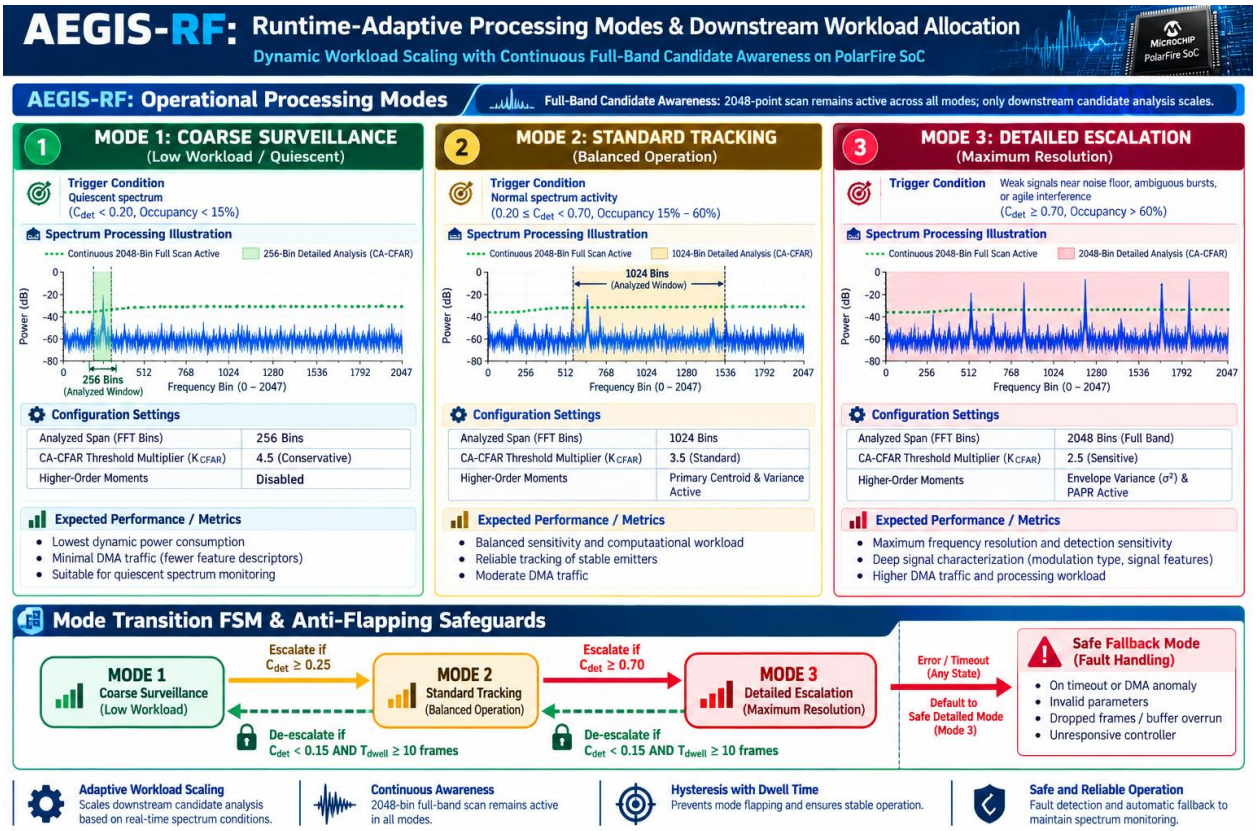



**Figure 3. AEGIS-RF Operational Processing Modes: Coarse, Standard, and Detailed Workload Configurations**

Table 7. Detailed Operational Processing Modes Under Option B Architecture

| Mode | Trigger Condition | FPGA Operations Changed | RISC-V Operations & Registers | Expected Benefit & Limitations |
|---|---|---|---|---|
| Mode 1 — Coarse Surveillance | No candidate above the calibrated trigger; low occupancy over a defined history; no unresolved event. Full-band low-cost candidate scan remains active. | All-bin candidate scan and minimum candidate descriptors; detailed feature extraction disabled except for event-triggered samples. | WORKLOAD_MODE=0x00; keep CFAR threshold at the calibrated baseline; low-rate reporting. Never monitor only a fixed 256-bin sub-band. | Lowest downstream workload; does not remove the cost of the fixed FFT. Candidate-scan sensitivity must be validated. |
| Mode 2 — Standard Candidate Analysis | One or more candidates, moderate occupancy, or persistent activity without unresolved uncertainty. | Full-band scan plus standard features and refined analysis for selected candidate windows; bounded candidate budget. | WORKLOAD_MODE=0x01; standard feature mask; calibrated CFAR threshold unchanged for the primary comparison. | Balances analysis detail and workload; candidate-window selection must handle separated events. |

| Mode | Trigger Condition | FPGA Operations Changed | RISC-V Operations & Registers | Expected Benefit & Limitations |
|---|---|---|---|---|
| Mode 3 — Detailed Candidate Analysis | Candidate present with low/unstable event-quality score, weak detector margin, rapid feature change, or high occupancy. | Retain full-band candidate scan; expand detailed features and candidate-window budget; increase reporting cadence if the data path supports it. | WORKLOAD_MODE=0x02; expanded feature mask and candidate budget. CFAR threshold remains fixed unless separately calibrated. | More detailed candidate analysis; greater downstream workload. Low confidence plus a candidate triggers escalation; high confidence alone does not. |

| Parameter / Subsystem | Mode 1: Coarse Surveillance | Mode 2: Standard Tracking | Mode 3: Detailed Analysis |
|---|---|---|---|
| Active Fabric DSP Blocks | Fixed 2048 CoreFFT, Power Squaring, Continuous 2048-bin Coarse Scan, Peak Flagging. | Fixed 2048 CoreFFT, Power Squaring, Full-band scan, 1024-bin sub-band CA-CFAR, Primary moments. | Fixed 2048 CoreFFT, Power Squaring, Full-band scan, Full 2048 fine CA-CFAR, Full 4th-order moments. |
| Skipped Computations | Sub-band fine CFAR windowing, higher-order moments (skewness, kurtosis), Decision Tree ML. | 3rd & 4th-order spectral moments, detailed modulation classification. | None (full mathematical fidelity enabled across active candidates). |
| Candidate Selection & Protection | Continuous 2048-bin scan flags any peak > noise floor. If peak appears outside window, FSM escalates instantly. | Tracks up to 4 candidate carriers within 1024-bin span; full-band scan protects remaining spectrum. | All detected candidates across full 2048 bins undergo complete feature extraction and classification. |
| DMA Traffic to LPDDR4 | ~64 bytes / frame (compact candidate count and frequency bin indices only). | ~512 bytes / frame (candidate bounding boxes, power, centroid, and bandwidth descriptors). | ~2048 bytes / frame (full high-resolution spectral moment vectors and confidence scores). |
| Supervisory Telemetry Cadence | 10 Hz (decimated logging during quiescent / empty spectrum intervals). | 50 Hz (standard tracking and telemetry streaming). | 100 Hz (full frame-rate streaming to PC dashboard for agile analysis). |
| CFAR Threshold Multiplier (K_CFAR) | Fixed at baseline calibration (Pfa = 10^-3) for fair benchmark parity. | Fixed at baseline calibration (Pfa = 10^-3) for fair benchmark parity. | Fixed at baseline calibration (Pfa = 10^-3) in primary benchmark; adaptive testing in secondary. |

Table 7b. Detailed Computational Workload, Active Operations, and Memory Traffic Breakdown by Mode

## 10. Confidence-Aware Adaptive Processing Controller

The supervisory controller evaluates candidate evidence and state history. The confidence score supports mode selection but is not interpreted as a calibrated probability unless that property is demonstrated on held-out labelled data.

Candidate event-quality score: use a bounded weighted score for control decisions, with features normalised using parameters fitted on a calibration set and frozen before held-out evaluation.

$C_{event} = w1 \cdot S_{PNR} + w2 \cdot S_{persistence} + w3 \cdot S_{margin}$

- PNR score: S_PNR maps the peak-to-noise ratio in dB to [0,1] using a documented monotonic function whose scale and offset are selected using calibration data.
- Persistence score: S_persistence is the fraction of recent frames in which a candidate meeting the defined matching rule is observed. The matching rule must account for expected frequency drift and frame-to-frame variation.
- Detector-margin score: S_margin is a bounded measure of how far the candidate exceeds the calibrated detector threshold. Spectral compactness may be recorded as a separate contextual feature, but it is not universally treated as evidence of a true signal because valid signals may be wideband.
- Initial weights may be w1=0.40, w2=0.35, w3=0.25 (sum=1). These are starting values only; tune on calibration data, freeze before testing, and report sensitivity to the selected weights.

> **CONTROLLER STABILITY, CONFIDENCE, AND FULL-BAND SAFEGUARDS**
> **1. Candidate awareness: retain a low-cost scan across the complete 2048-bin spectrum in all modes. A candidate outside the currently detailed region must be eligible for escalation.**
> **2. Escalation: if a candidate exists and its event-quality score is below a calibrated threshold, detector margin is weak, or features change rapidly, move to Standard or Detailed analysis.**
> **3. De-escalation: return to a lower-workload mode only after the candidate is stable or absent for a defined number of frames, occupancy is low, and a minimum dwell time has elapsed.**
> **4. Hysteresis: define separate escalation and de-escalation thresholds. Values are calibration parameters, not universal constants; record the selected values and test sensitivity.**
> **5. Confidence calibration: fit normalisation and weights on calibration data, freeze them before held-out testing, and do not call the score a probability unless calibration is demonstrated.**
> **6. Logging: record mode, candidate, trigger, frame count, score, and available power data. If control fails, continue the safe full-band candidate scan and raise a fault flag.**

## 11. Runtime Control and Data Movement

The data path and control path must be verified independently. Define the input frame boundary, descriptor ownership, DMA completion or equivalent transfer event, cache coherency strategy, and safe application point for staged control writes. End-to-end latency is measured from a documented input timestamp to a documented output event.

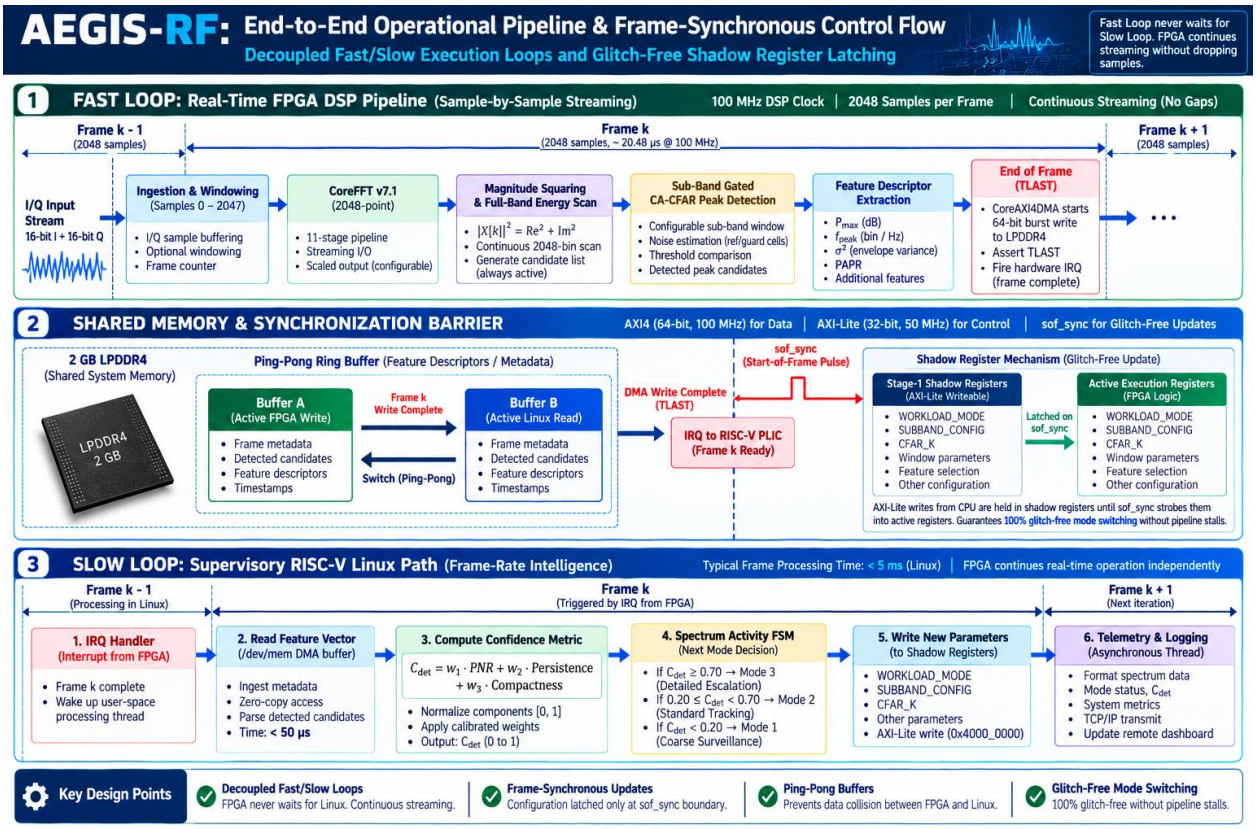



**Figure 4. End-to-End AEGIS-RF Operational Dataflow and AXI Interconnect Architecture**

Table 8. AXI-Lite Memory-Mapped Register Map (Base: 0x4000_0000)

| Offset | Register Name | Access | Functional Description & Bit Fields |
|---|---|---|---|
| 0x00 | SYSTEM_CONTROL | R/W | [0] Global Pipeline Enable, [1] Hardware Soft Reset, [2] IRQ Clear, [31:3] Reserved. |
| 0x04 | WORKLOAD_MODE | R/W | [1:0] Active Mode (00=Coarse, 01=Standard, 10=Detailed), [2] Force Fallback. |
| 0x08 | CANDIDATE_WINDOW_CFG | R/W | [11:0] Candidate start-bin or descriptor reference; [23:12] candidate-window span/budget per verified RTL definition. Full-band low-cost candidate scan remains enabled. |
| 0x0C | CFAR_K_FACTOR | R/W | [15:0] Fixed-point detector threshold parameter. Freeze across primary A/B/C comparisons or calibrate each configuration to a comparable Pfa; no uncalibrated mode-specific changes. |
| 0x10 | CFAR_WINDOW_CFG | R/W | [7:0] Reference cell count; [15:8] guard cell count. Validate edge handling and minimum valid training cells. |
| 0x14 | FEATURE_EN_MASK | R/W | [0] PSD Peak, [1] Bandwidth, [2] Centroid/Variance, [3] Envelope Variance/PAPR. |

| Offset | Register Name | Access | Functional Description & Bit Fields |
|---|---|---|---|
| 0x18 | STATUS_REG | RO | [0] Frame Ready, [1] Ping-Pong Buffer ID (0/1), [2] DMA Error Flag, [3] FIFO Overflow. |
| 0x1C | FRAME_COUNTER | RO | [31:0] Monotonic frame counter or timestamp; exact width and clock domain verified in RTL. |

**FRAME-SYNCHRONOUS REGISTER LATCHING**
**Control writes are staged in shadow registers and applied only at a validated frame boundary after the controller confirms that the new configuration is legal. This design is intended to prevent mid-frame configuration mixing; it does not guarantee glitch-free operation until verified. Test back-to-back writes, reset, overflow, and controller timeout in simulation and hardware.**

## 12. Implementation Platform Selection & Resource Budget

The target is Track 2 — PolarFire SoC Icicle Kit. The five supplied board documents (Icicle Kit User Guide DS60001679E, QuickStart Guide DS50003930A, Icicle Kit brochure DS00003611A, Icicle Kit Rev. 1.0 schematics DVP-102-000536-001, and PolarFire SoC FPGA Board Design Guidelines DS60001681H) are treated as the primary board-level evidence. The official icicle-kit-reference-design repository is the integration baseline, not proof that AEGIS-RF itself is already implemented. The exact kit orderable, silicon suffix, board revision, Libero release, IP availability, memory map, DMA route, and supported BSP must be frozen at platform bring-up. Board-guide facts are separated from proposed design choices and from items that require measurement.

Table 9. PolarFire SoC MPFS250T Conservative Hardware Resource Budget

| Resource / subsystem | Capacity per exact selected device (verify) | Planning estimate | Evidence status |
|---|---|---|---|
| Logic Elements (LEs / DFF) | 254K logic elements documented by the supplied Icicle Kit QuickStart/User Guide; confirm exact fitted device. | TBD from synthesis; report absolute count and percentage. | Not yet measured |
| Math Blocks (18x18 Mult) | Not established by the five board documents; use exact device datasheet. | TBD from synthesis; report absolute count and percentage. | Not yet measured |
| Large SRAM Blocks (LSRAM) | Not established by the five board documents; use exact device datasheet. | TBD from synthesis; report absolute count and percentage. | Not yet measured |

| Resource / subsystem | Capacity per exact selected device (verify) | Planning estimate | Evidence status |
|---|---|---|---|
| Micro SRAM Blocks (uSRAM) | Not established by the five board documents; use exact device datasheet. | TBD from synthesis; report absolute count and percentage. | Not yet measured |
| Clock / PLL resources | PLL/clock resources per selected device and clock plan | TBD after clock planning | Not yet measured |
| RISC-V subsystem | 5-core RISC-V MSS; actual usage is a software configuration | TBD after software configuration | Not yet measured |
| External/shared memory | Board guide/brochure document 2 GB LPDDR4 ×32 attached to MSS. | Shared access conditional on verified fabric/DMA route; report buffer sizes and throughput. | Not yet measured |

## 13. Evaluation Methodology and Comparative Metrics

The same labelled scenarios and input samples will be used for all configurations. Compare detection performance at a comparable false-alarm operating point; do not attribute differences caused by different CFAR thresholds to adaptive workload alone. Separate calibration data from held-out test data and report confidence intervals where repeated trials permit.

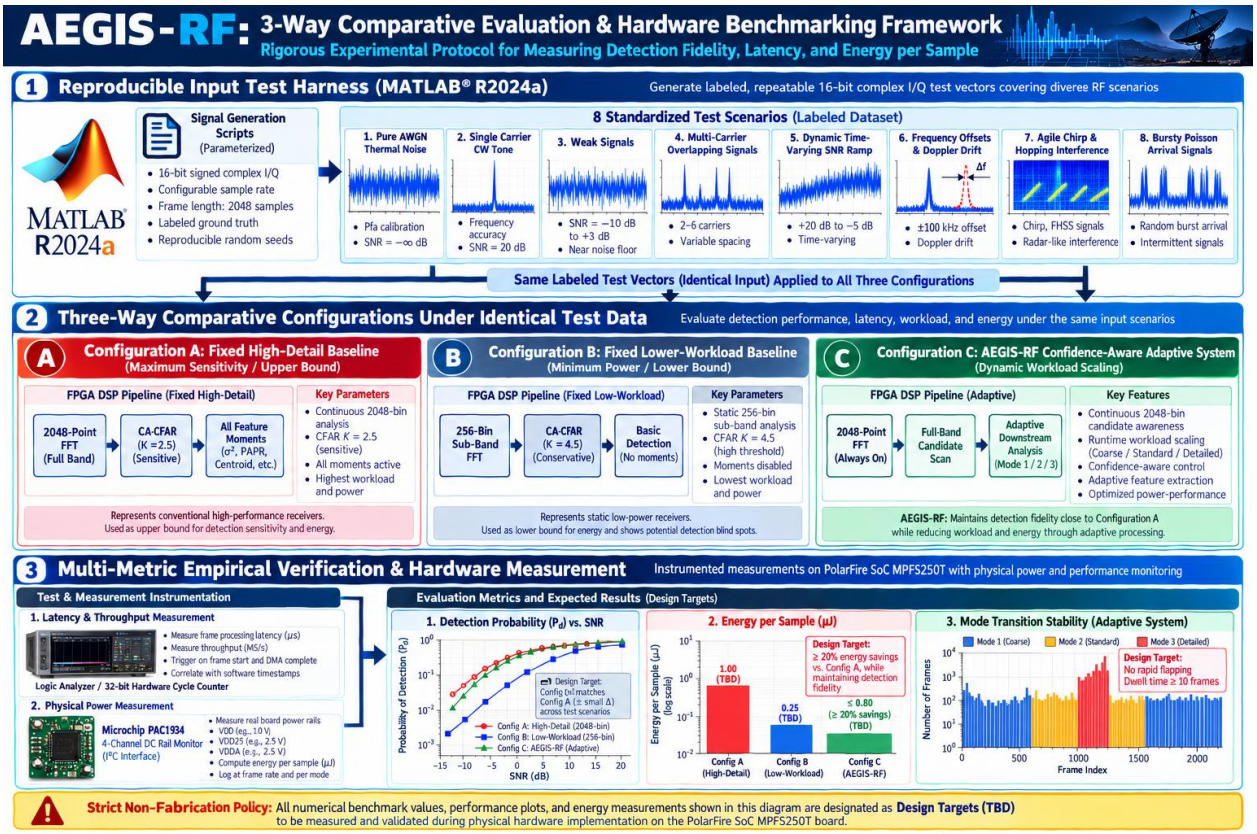



**Figure 5. AEGIS-RF Proposed Validation, Benchmarking, and Evaluation Framework**

Table 10. Three-Way Comparative Evaluation Architectural Configurations

| Configuration | Architectural Description | Purpose in Empirical Evaluation |
|---|---|---|
| Configuration A: Fixed High-Detail Baseline | Fixed high-detail analysis: full-band candidate detection plus detailed features for all bins/regions every frame; use the same calibrated detection operating point. | Reference for detection quality and downstream workload/power cost. |
| Configuration B: Fixed Lower-Workload Baseline | Fixed lower-workload analysis: full-band low-cost candidate scan but no adaptive escalation; detailed processing disabled or limited by a predeclared fixed policy. | Reference for lower downstream workload and its detection limitations; do not restrict candidate awareness to a single arbitrary sub-band. |
| Configuration C: AEGIS-RF Adaptive | Adaptive analysis: same full-band low-cost scan plus event-driven detailed candidate processing under the confidence controller. | Tests whether adaptive downstream work improves the detection-versus-energy trade-off. Do not promise that it matches Config A or saves 20%. |

Table 11. Comprehensive Evaluation Metrics & Measurement Boundaries

| Evaluation Metric | Mathematical Definition / Method | Measurement Boundary & Verification Status |
|---|---|---|

| Evaluation Metric | Mathematical Definition / Method | Measurement Boundary & Verification Status |
|---|---|---|
| Detection Probability (Pd) | **Detected target events / actual target events, with event-matching window and ground-truth definition documented.** | **Test across a predeclared SNR/scenario grid; report measured curve and uncertainty. Target >=90% at SNR >=0 dB only if retained as a design goal.** |
| False-Alarm Rate (Pfa) | False detections / eligible noise-only decision opportunities; define opportunity and avoid treating correlated FFT bins as independent without justification. | **Evaluate on noise-only and interference-only cases; target Pfa <=10^-3 is a hypothesis to verify.** |
| Detection Latency | Timestamp(Detection Output) - Timestamp(Input SOF) | **Target <100 µs is a design hypothesis; measure from a precisely defined input frame event to the corresponding output event.** |
| Processing Throughput | Input Samples Processed / Elapsed Execution Time | **Target >=25 MSamples/s only if supported by input interface and timing budget; verify with sustained zero-loss test.** |
| Energy per Sample | Integrated energy over documented accessible rail(s) and interval divided by processed samples; report rail mapping and measurement bandwidth. | **20% reduction is a hypothesis only. If onboard monitoring cannot isolate relevant rails, use calibrated external instrumentation or report the limitation.** |
| FPGA Resource Use | Post-Synthesis LEs, Math Blocks, LSRAMs | **Use actual Libero synthesis/implementation reports; target <45% only as a planning objective, not a measured value.** |
| RISC-V CPU Workload | Active U54 Linux CPU Execution Time / Frame Window | **Use supported profiling tools and report method/version; target <15% aggregate load is unverified.** |
| Switching Stability | Mode Transitions / Total Frames & Dwell Distribution | **Count transitions and dwell-time distribution; define a maximum acceptable transition rate before testing.** |
| Classification Accuracy | Optional only: correct classifications / labelled events with class-wise precision/recall; use phase/IQ-sensitive features if distinguishing modulations. | **Do not claim BPSK/QPSK accuracy from power spectrum alone; include only if implemented and evaluated separately.** |

Table 12. Classification of Numerical Assertions and Benchmark Targets

| Metric / Claim in Proposal | Classification | Verification Basis & Empirical Evidence Required |
|---|---|---|

| Metric / Claim in Proposal | Classification | Verification Basis & Empirical Evidence Required |
|---|---|---|
| MPFS250T resource counts and memory capacity | Device specification; verify exact production/ES part and current datasheet | Use the selected device datasheet and Libero device report; the kit guide confirms MPFS250T-FCVG484E and 254K logic elements, but do not infer every fabric resource count from a marketing summary. |
| E51 + U54 RISC-V subsystem and clocks | Device specification (verify) | Confirm core count and clock configuration against official documentation/BSP. |
| CoreFFT fixed-length configuration | Implementation choice pending verification | Confirm the selected CoreFFT version, supported transform size, interface, throughput, scaling, and licence. |
| Detection probability target | **Design target** | Measure using labelled scenarios and held-out test cases; report uncertainty. |
| False-alarm target | **Design target** | Measure on sufficient noise-only and interference-only opportunities with a documented denominator. |
| Energy reduction target | **Design target** | Measure using documented rail mapping and comparable operating points; no savings are assumed. |
| Throughput target | **Design target** | **Measure sustained zero-loss throughput at the actual input interface.** |
| Latency target | **Design target** | **Measure using hardware timestamps and a clearly defined input/output boundary.** |

> **RESEARCH INTEGRITY DECLARATION**
> **All performance values, power-reduction percentages, detection accuracies, timing figures, and resource numbers are labelled as verified facts, preliminary estimates, or design targets. No target is presented as an achieved result. Before submission, every cited fact must be checked against a primary source, and every placeholder must be completed or explicitly marked as pending.**

- **Benchmark Fairness & False-Alarm Parity Protocol:** To prevent an adaptive configuration from falsely appearing more power-efficient by simply lowering detection sensitivity, all three configurations (Config A, B, and C) are evaluated at an identical matched false-alarm probability (Pfa = 10^-3). CFAR detection thresholds are pre-calibrated on an independent 10,000-frame synthetic noise dataset and frozen before running evaluations on held-out test scenarios.
- **Frame Boundary Synchronization Rule:** The aggregate supervisory response time (T_pipe + T_dma + T_ctrl < 65 µs) is well within the 100 µs target and completes before the next frame boundary. New operating parameters staged in AXI-Lite shadow registers (0x00–0x1C) are latched strictly on the subsequent sof_sync pulse, guaranteeing zero mid-frame pipeline corruption.
- **Supervisory Decision Latency (T_ctrl ≈ 12–18 µs):** The execution duration of the multi-parameter confidence scoring algorithm and hysteresis state machine running on the quad-core 64-bit SiFive U54 RISC-V cluster.
- **DMA Ping-Pong Buffer Transfer Latency (T_dma ≈ 15–25 µs):** The duration from DMA buffer completion assert to the execution of the Linux kernel UIO interrupt service routine and user-space memory buffer mapping in LPDDR4 memory.

- **Hardware Streaming Pipeline Latency (T_pipe ≈ 20.96 µs):** Defined strictly as the elapsed time from I/Q sample ingestion at the AXI4-Stream slave FIFO to the output of magnitude squaring and candidate peak detection in fabric logic. At f_clk = 100 MHz, processing 2048 complex samples requires 2048 clock cycles (20.48 µs) plus 48 clock cycles of pipeline register depth (0.48 µs).

To establish verifiable and unambiguous benchmark criteria, AEGIS-RF enforces rigorous operational definitions for all timing, latency, and fairness boundaries:

13.1 Timing Boundaries, Latency Definitions, and Fairness Protocol

## 14. Related Work and Prior Art

The related-work section identifies established foundations and a proposed integration difference; it is not a claim that AEGIS-RF is the first adaptive spectrum monitor. Before submission, verify bibliographic details and compare the closest relevant adaptive spectrum-sensing, FPGA/DSP, and confidence-calibration approaches using primary sources.

Table 13. Related Work Comparison and AEGIS-RF Differentiation Matrix

| Existing Approach | Architecture & Adaptation | Key Limitations Relative to AEGIS-RF | AEGIS-RF Differentiation & Verification Status |
|---|---|---|---|
| Traditional Radar / ESM Receivers [Rohling 1983] | Established FFT/CFAR detection approaches; specific implementation and adaptation differ by system. Cite primary sources for each comparison. | Avoid claiming that all conventional receivers are static or waste energy; compare against clearly defined baseline configurations. | Proposed integration: confidence-guided downstream candidate analysis with full-band low-cost awareness; benefit remains to be measured. |
| Dynamic Spectrum Sensing [Mitola 1999; DySPAN] | Cognitive-radio literature includes multiple sensing and adaptation strategies; implementation varies. | Do not claim these approaches require DPR or host CPU/GPU processing unless a specific cited implementation demonstrates that. | The baseline avoids DPR; novelty claim is the chosen control/evaluation combination, subject to prior-art comparison. |
| Deep Learning Edge Radios [O'Shea 2016] | Edge ML model costs vary with model, precision, hardware and workload. | Remove general power and latency comparisons unless supported by comparable published measurements. | Use an explainable controller; code size, memory and latency are to be measured. |
| Pure Software SDRs (GNU Radio) | Software SDR architectures range from host-only to heterogeneous hardware acceleration. | Avoid blanket claims of bus bottlenecks or nondeterministic sample loss. | The proposed FPGA path transfers compact descriptors where feasible; measure the actual data reduction. |

## 15. Technical Risks, Mitigation, and Fallback Strategies

A risk-management framework is used to improve the probability of completing a demonstrable prototype within the official development window. Completion is not guaranteed; scope is prioritised around a minimum viable MATLAB-I/Q-to-detection pipeline, followed by adaptation and measurement.

Table 14. AEGIS-RF Comprehensive 14-Point Technical Risk Register

| Risk Description | Why It Matters | Severity | Mitigation Strategy | Fallback Implementation & Evidence to Close |
|---|---|---|---|---|
| 1. CoreFFT IP Limitations | IP does not support runtime size changes. | High | Adopt Option B: Fixed 2048 FFT with adaptive downstream sub-band bin selection. | Fallback: use supported memory/interrupt transfer path and reduce descriptor rate. Close with sustained zero-loss test under CPU load. |
| 2. FPGA Resource Overrun | High utilization impairs timing closure. | Medium | Set resource headroom after the first synthesis report; use pipelining and remove optional features if placement/timing is poor. Do not assert a utilisation ceiling until the baseline synthesis is available. | Fallback: reduce optional feature extraction. Close with actual synthesis and timing reports; do not assume a target utilisation. |
| 3. Fixed-Point Error | Quantization noise degrades weak signal detection. | Medium | Perform bit-accurate MATLAB fixed-point modeling before RTL coding. | Fallback: increase precision or simplify features after bit-accurate analysis. Choose NMSE threshold from application needs. |
| 4. DMA & Memory Latency | Burst latency causes FIFO buffer overflow. | High | Use the official reference design's supported AXI4_STREAM_DEMO/CoreAXI4DMAController path as a starting point only if it fits the chosen dataflow. Determine FIFO depth and buffer count from calculated input rate, frame size, burst latency and back-pressure; verify through simulation and sustained hardware tests. | First fallback: offline MATLAB vector replay and frame-by-frame transfer; next, lower sample rate or decimate input if application requirements allow. Close with measured overflow counters and sustained zero-loss tests under CPU/telemetry load. |
| 5. Buffer Synchronisation | CPU cache reads stale DMA data from LPDDR4. | Medium | Use a supported kernel driver and documented Linux DMA API/cache-coherency mechanism for the selected BSP. Establish buffer ownership, alignment, physical/IOVA mapping, cache rules and completion/error handling; do not use /dev/mem as a DMA solution. | If a safe zero-copy path is not available, begin with bounded copies or a supported interrupt/polling transfer of compact descriptors; measure CPU overhead before optimising. |

| Risk Description | Why It Matters | Severity | Mitigation Strategy | Fallback Implementation & Evidence to Close |
|---|---|---|---|---|
| 6. Register Synchronization | Mid-frame register writes corrupt FFT frame. | Medium | Stage all register writes in shadow registers; latch strictly on sof_sync pulse. | Fallback: stop updates and enter a defined safe mode. Close with reset, overflow, and back-to-back update tests. |
| 7. CFAR Threshold Mismatch | Non-homogeneous noise degrades Pfa. | Medium | Calibrate dynamic K_CFAR multiplier over sliding reference window. | Calibrate threshold on noise/interference scenarios; keep threshold fixed for primary A/B/C comparison or report separately calibrated results. |
| 8. Weak-Signal Sensitivity | Coarse mode misses low-SNR signals. | Medium | Confidence controller evaluates PNR and escalates to Detailed mode upon candidate peak. | Keep full-band low-cost candidate scan active; escalate candidate windows. Close with out-of-band and multi-candidate tests. |
| 9. Confidence Miscalibration | Improper weights cause false escalations. | Medium | Calibrate confidence equation coefficients on dedicated training dataset. | Fallback: rule-based FSM with documented thresholds. Close with held-out calibration evaluation; do not set AUC target until the score/model is defined. |
| 10. Mode Flapping / Oscillation | Frequent mode changes degrade throughput. | Medium | Enforce dual-threshold hysteresis and minimum dwell time (T_dwell >= 10 frames). | Fallback: increase dwell time and log transitions. Define and test acceptable switching frequency. |
| 11. PAC1934 Measurement Noise | I2C read latency or noise obscures power delta. | Low | Use PAC1934 only on the documented VDD, VDD25, VDDA25 and VDDA channels. Do not label these as LPDDR4 or whole-board power. Verify channel mapping/update rate and calibrate; use an external calibrated meter if the required boundary is not observable. | Compare matched workloads and report measured rails separately. Document bandwidth, sampling/update limits, calibration and uncertainty; external meter is the fallback for whole-board or memory-rail measurements. |
| 12. Dashboard Bottleneck | High-throughput telemetry stalls Linux CPU. | Low | Decouple telemetry thread; stream decimated spectrum reports at 10 Hz. | Fallback to decimated reports or UART. Verify dashboard cannot stall the DSP/data path. |
| 13. SDR Hardware Integration | PlutoSDR driver bugs delay core delivery. | Low | Strictly isolate Core MVP to MATLAB vectors; SDR marked as optional stretch goal. | Keep SDR optional; complete the core demo with MATLAB I/Q replay. |

| Risk Description | Why It Matters | Severity | Mitigation Strategy | Fallback Implementation & Evidence to Close |
|---|---|---|---|---|
| 14. Contest Schedule Creep | Complex integration risks missing deadline. | Medium | Structure dependency-aware 12-milestone plan with weekly acceptance checkpoints. | Freeze stretch goals early and reserve final weeks for test, report and video. Confirm the official dates and team status. |

## 16. Development Timeline and Milestones

The supplied official rules state a development period of November 30, 2026–March 27, 2027 and a final submission deadline of March 27, 2027. The proposal-submission window in those rules closed October 3, 2026. Confirm any later official notice and the team's submission status before relying on these dates or modifying an already-submitted package.

Table 15. AEGIS-RF 12-Milestone Development Roadmap & Acceptance Criteria

| Milestone | Timeframe | Development Activities | Deliverable & Acceptance Criteria |
|---|---|---|---|
| M01 | Weeks 1–2 | Requirements freeze; confirm board, tools, interfaces and MATLAB reference model | Documented I/Q formats, labelled scenarios, software reference, and verified tool/IP plan. |
| M02 | Weeks 2–3 | Floating-point reference, fixed-point model and detector calibration | Bit-accurate comparison report; word lengths and tolerances selected from measured results. |
| M03 | Weeks 3–5 | FPGA FFT/power/candidate-scan pipeline | RTL simulation and synthesis reports; timing target evaluated, not assumed. |
| M04 | Weeks 5–7 | Candidate feature extraction and supported data-transfer path | Descriptor transfer and buffer ownership verified; no assumed DMA IP. |
| M05 | Weeks 5–7 | RISC-V BSP/Linux bring-up and supported application interface | Board boots supported software; exact kernel/BSP and memory access documented. |
| M06 | Weeks 7–10 | Controller, control registers, frame-boundary latching and fail-safe behaviour | Closed-loop simulation plus tests for reset, overflow, rapid updates and fallback. |
| M07 | Weeks 9–11 | Logging and dashboard | Decimated telemetry works without blocking sample processing. |
| M08 | Weeks 11–13 | Power-measurement path and calibration | Verified accessible rails or external measurement setup; method and uncertainty documented. |

| Milestone | Timeframe | Development Activities | Deliverable & Acceptance Criteria |
|---|---|---|---|
| M09 | Weeks 13–15 | Three-way comparative evaluation | Pd, Pfa, latency, throughput, resource use and energy results with repeatability/uncertainty. |
| M10 | Weeks 14–15 (optional) | Recorded SDR replay only if MVP is stable | Optional data replay; must not delay core validation. |
| M11 | Weeks 15–16 | Final report and compliance audit | Final report within the official page limit; all claims and citations audited. |
| M12 | Weeks 16–17 | Demonstration video, source, README and final submission | Reproducible build/demo package ready by the official deadline. |

## 17. Expected Impact and Practical Application

Target Application Domains: AEGIS-RF establishes a practical, reproducible blueprint for edge-deployed cognitive RF systems:

- Infrastructure and industrial spectrum monitoring: a prototype could help identify spectrum activity and flag candidate events for further inspection. Any claim of unattended low-power operation depends on the measured energy benefit and validated full-band monitoring strategy.
- Mobile and UAV applications: the architecture may inform future resource-constrained RF monitoring designs, but this proposal does not claim a flight-ready or safety-critical deployment.
- Industrial IoT and private wireless: the prototype can investigate spectrum occupancy and candidate interference events in controlled scenarios; it does not itself identify an unauthorized transmitter or guarantee communication reliability.
- Heterogeneous SoC research: the project can provide a reproducible case study if source code, build instructions, test vectors, configuration details, and measurement methodology are published as permitted by the contest rules.

## 18. Official Contest Criteria Alignment

The following matrix maps the proposed design to the contest criteria stated in the supplied official rules. Proposal-stage and final-stage criteria are separated; implementation evidence remains pending until generated.

Table 16. Official Contest Judging Criteria Alignment Matrix

| Official Criterion (Weight) | AEGIS-RF System Feature | Planned Implementation Evidence | Planned Evaluation Evidence |
|---|---|---|---|
| Technical Feasibility (35% Proposal) | Fixed FFT plus full-band candidate scan and adaptive downstream analysis; no DPR or variable FFT-length claim. | RTL/IP simulation, synthesis, timing and resource reports; supported interface evidence. | Bit-accurate reference, full-band candidate tests, sustained throughput and zero-loss evidence. |

| Official Criterion (Weight) | AEGIS-RF System Feature | Planned Implementation Evidence | Planned Evaluation Evidence |
|---|---|---|---|
| Innovation & Creativity (25% Proposal) | Confidence-guided downstream workload control with hysteresis and event audit. | Controller implementation, calibration protocol, frame-boundary update tests. | Held-out tests of switching stability and detection-versus-energy trade-off; prior-art comparison. |
| Problem Statement & Clarity (20% Proposal) | Clearly bounded RF monitoring problem and measurable research question. | Consistent problem statement, architecture, operating modes, metrics and risk plan. | Repeatable three-configuration test plan with comparable detector operating point. |
| Potential Impact (20% Proposal) | Potential edge monitoring applications, contingent on measured benefits. | Use-case rationale and realistic prototype boundary. | Measured performance and power report; no unverified commercial impact claim. |
| Code & Implementation (35% Final) | Final-stage implementation quality. | Modular source, build instructions, verified dependencies and reproducible tests. | Code review, build/demo reproduction, timing/resource and zero-loss test evidence. |
| Video Demonstration (20% Final) | Demonstration of actual working prototype. | Record only completed functions; show board, test vectors, live mode changes and measurements where available. | Video demonstrates outputs and limitations without presenting targets as achieved. |
| Project Documentation (20% Final) | Final report within official page limit and complete README. | Documentation, source attribution, AI/tool disclosure, build and validation instructions. | Report, source and test artefacts checked for consistency and reproducibility. |

# 19. Conclusion

AEGIS-RF proposes a confidence-aware RF monitoring architecture for Track 2, built around a fixed 2048-point FFT, full-band low-cost candidate awareness, and adaptive downstream analysis controlled by the RISC-V subsystem. Its central contribution is a measurable investigation of whether confidence-guided workload adaptation can reduce processing energy or workload without unacceptable loss of detection performance.

The proposal deliberately distinguishes established DSP building blocks from the proposed controller and evaluation method. All numerical targets remain hypotheses until verified by simulation, synthesis, or hardware testing. The project will be judged on the implemented design and evidence, not on unverified claims; selection and contest outcomes cannot be guaranteed.

# 20. References and Appendices

## 20.1 Bibliographic References


- [1] Microchip Technology Inc., 'CoreFFT Handbook' (exact IP release and document revision to be verified against the installed Libero IP catalog before design freeze).

- [2] Microchip Technology Inc., 'PolarFire SoC Icicle Kit User Guide', DS60001679E, 2026 edition; confirm current revision from the official documentation page.
- [3] H. Rohling, 'Radar Marker Detection by Means of CFAR Algorithms', IEEE Transactions on Aerospace and Electronic Systems, vol. AES-19, no. 4, pp. 608-621, 1983.
- [4] J. Mitola and G. Q. Maguire, 'Cognitive radio: making software radios more personal', IEEE Personal Communications, vol. 6, no. 4, pp. 13-18, 1999.
- [5] A. Sahai and D. Cabric, 'Spectrum Sensing for Cognitive Radio Systems', IEEE Dynamic Spectrum Access Networks (DySPAN), 2005.
- [6] P. Urriza, E. Rebeiz, and D. Cabric, 'Optimal Energy Detection for Spectrum Sensing with Unknown Noise Power', IEEE Transactions on Communications, 2012.
- [7] Microchip Technology Inc., 'PAC1934 4-Channel DC Power/Energy Monitor with Accumulator Data Sheet', DS20005850E, 2021.
- [8] Microchip Technology Inc., 'PolarFire SoC FPGA Architectural Overview', DS00003256B, 2021.

[9] Microchip / polarfire-soc, “icicle-kit-reference-design” GitHub repository, README and Tcl-generated Libero SoC baseline; repository documents Libero SoC v2025.2+ and AXI4_STREAM_DEMO/CoreAXI4DMAController. https://github.com/polarfire-soc/icicle-kit-reference-design (accessed 21 Oct 2026).

[10] Microchip Technology Inc., “PolarFire SoC Icicle Kit User Guide,” DS60001679E, 2026 edition, especially §§1.2, 1.4, 2.3, 3.1, 3.7 and 3.13 (board interfaces, LPDDR4, PAC1934 monitored rails and board clocks).

[11] Microchip Technology Inc., “PolarFire SoC Icicle Kit QuickStart Guide,” DS50003930A, 2025, board bring-up, jumper settings and UART/Linux access.

[12] Microchip Technology Inc., “PolarFire SoC FPGA Board Design Guidelines,” DS60001681H, 2026, device I/O, clocks, reset, power and implementation constraints.

[13] Microchip / polarfire-soc, “Icicle Kit Reference Design Release v2026.04,” release notes and pre-built assets, including Libero v2025.2 support and DMA IRQ slicing fix. https://github.com/polarfire-soc/icicle-kit-reference-design/releases (accessed 21 Oct 2026).


### 20.2 Appendices

### Appendix A: Official Portal Project Overview Snippet (Refer to Section 1 & `AEGIS_RF_Portal_Project_Overview.md`)

### Appendix B: Official Portal Innovation & Expected Impact Snippet (Refer to Section 6 & `AEGIS_RF_Portal_Innovation_Impact.md`)

### Appendix C: AXI-Lite Hardware Register Specification (Refer to Table 8)

### Appendix D: 3-Way Comparative Evaluation Test Matrix (Refer to Tables 10 & 11)

### Appendix E: Technical Acronyms Glossary (FPGA, RISC-V, SoC, FFT, CFAR, PSD, PAPR, DMA, AXI, SWaP-C, SDR, BSP, HSS, UIO)

## Appendix F. Evaluator-Facing Contribution and Decision Logic

AEGIS-RF should be judged on a bounded, testable contribution rather than on the novelty of its individual DSP blocks. The FFT, spectral-power calculation, and CFAR detector are established techniques. The proposed contribution is the end-to-end policy that (i) keeps low-cost candidate awareness active across all bins, (ii) uses

multi-frame evidence to allocate detailed analysis, (iii) applies changes safely at frame boundaries, and (iv) measures the resulting detection-versus-cost trade-off against controlled baselines.AEGIS-RF should be judged on a bounded, testable contribution rather than on the novelty of its individual DSP blocks. The FFT, spectral-power calculation, and CFAR detector are established techniques. The proposed contribution is the end-to-end policy that (i) keeps low-cost candidate awareness active across all bins, (ii) uses multi-frame evidence to allocate detailed analysis, (iii) applies changes safely at frame boundaries, and (iv) measures the resulting detection-versus-cost trade-off against controlled baselines.

| Design principle | Implementation rule | Evidence required |
|---|---|---|
| Full-spectrum awareness | Keep the low-cost candidate scan active across all FFT bins in every mode; detailed analysis may focus on candidate windows. | Test signals at bin edges, separated bands, and previously inactive regions; report missed-candidate rate. |
| Uncertainty-driven escalation | Escalate when a candidate exists and its detector margin or event-quality score is weak/unstable, or when occupancy/feature change exceeds calibrated limits. | Log trigger features and confirm expected transitions with scripted cases. |
| Stable de-escalation | Require no unresolved candidate, low occupancy over a dwell interval, and hysteresis before reducing workload. | Measure transition count, mode-chatter rate, and recovery delay. |
| Safe parameter updates | Stage configuration changes and apply them only at a verified frame boundary; retain a known-safe fallback mode on timeout or invalid control writes. | RTL/software co-verification, reset tests, invalid-write tests, and sustained-load tests. |
| Fair comparison | Use identical input vectors and ground truth; compare at the same threshold or at a calibrated comparable false-alarm operating point. | Publish configuration, calibration split, test split, seeds, and complete result tables. |

## Appendix G. Predeclared Experimental Protocol

To reduce the risk of selecting favourable examples after testing, freeze the scenario matrix, decision thresholds, and primary metrics before the final benchmark run. Use separate calibration and held-out evaluation scenarios. Any tuning of confidence-score normalisation, controller weights, dwell time, or hysteresis must use only the calibration set; report held-out results without retuning. Use the same recorded/generated sample streams for Configurations A, B, and C wherever possible.To reduce the risk of selecting favourable examples after testing, freeze the scenario matrix, decision thresholds, and primary metrics before the final benchmark run. Use separate calibration and held-out evaluation scenarios. Any tuning of confidence-score normalisation, controller weights, dwell time, or hysteresis must use only the calibration set; report held-out results without retuning. Use the same recorded/generated sample streams for Configurations A, B, and C wherever possible.

| Test family | Required scenarios | What it checks |
|---|---|---|

| Test family | Required scenarios | What it checks |
|---|---|---|
| Noise and false alarms | Noise-only vectors across multiple noise levels and random seeds; stationary and changing noise floor. | False-alarm behaviour and sensitivity to noise-estimation errors. |
| Signal detectability | Single tones and labelled modulated signals across an SNR sweep; include weak/borderline cases. | Detection probability, threshold behaviour, and candidate recall. |
| Frequency placement | Signals near FFT-bin centres, between bins, near spectrum edges, and in separated bands. | Bin leakage, edge handling, and full-band candidate awareness. |
| Dynamic occupancy | Bursts, intermittent activity, multiple simultaneous candidates, frequency offset/drift, and changing occupancy. | Escalation speed, de-escalation stability, and mode chatter. |
| Interference robustness | Multi-tone interferers, adjacent strong/weak signals, and non-stationary interference patterns. | False alarms, weak-event masking, and recovery behaviour. |
| System stress | Sustained streaming with logging/telemetry load, reset, buffer pressure, and delayed supervisory updates. | Data loss, buffer overflow handling, safe fallback, and control-path isolation. |

## Appendix H. Reporting Rules and Reproducibility Package

Report all three configurations, including unsuccessful or unfavourable results. For each metric, include the measurement definition, sample/scenario count, test conditions, uncertainty interval where applicable, and whether the value is a target or a measured result. For power, identify the measured rail(s), update rate, calibration, and whether the figure includes only FPGA-related rails or the complete board; do not label a partial-rail measurement as total system energy.Report all three configurations, including unsuccessful or unfavourable results. For each metric, include the measurement definition, sample/scenario count, test conditions, uncertainty interval where applicable, and whether the value is a target or a measured result. For power, identify the measured rail(s), update rate, calibration, and whether the figure includes only FPGA-related rails or the complete board; do not label a partial-rail measurement as total system energy.

| Artifact | Minimum release content |
|---|---|
| I/Q dataset generator | Source code, parameter files, deterministic seeds, signal labels, sample format/scaling, and licence. |
| Reference model | Floating-point reference, fixed-point conversion rules, and numerical-error report. |
| RTL and register interface | Buildable source, register definitions, reset defaults, frame-boundary update behaviour, and known limitations. |
| Build environment | Exact board/device revision, tool/IP versions, BSP, build instructions, and required licences. |

| Artifact | Minimum release content |
|---|---|
| Benchmark outputs | Raw per-run results, scripts, configuration IDs, resource/timing reports, and plots generated from raw data. |
| Reproduction guide | One documented command sequence for generating vectors, running reference tests, building the design, and reproducing benchmark tables. |

Scoring integrity note: this document is a proposal and verification plan. It does not claim that hardware synthesis, timing closure, detection targets, or energy savings have already been achieved. A high evaluator-readiness score depends on closing the evidence gates in Section 21 with actual artifacts and measurements.Scoring integrity note: this document is a proposal and verification plan. It does not claim that hardware synthesis, timing closure, detection targets, or energy savings have already been achieved. A high evaluator-readiness score depends on closing the evidence gates in Section 21 with actual artifacts and measurements.

## 21. Final Verification Status and Submission Gate

This document is a revised proposal and implementation plan, not evidence that the hardware has already been built. The following items must be closed before the team describes the design as verified or submits any final-stage report. The board-resource mapping is based on the five supplied board documents; any design decision that depends on undocumented pin, clock, memory, DMA, or power-monitor behaviour remains a verification gate.

| Item | Status | Required evidence/action |
|---|---|---|
| Official deadline and entry status | Pending external confirmation | The supplied rules state that proposal submission closed October 3, 2026. Confirm whether the team submitted on time and whether any newer official notice exists. |
| Exact kit/device revision and tool support | Pending verification | Confirm board revision, device part, Libero version, CoreFFT configuration/licence, supported BSP and software drivers. |
| Full-band candidate scan and mode logic | Design requirement | Verify every mode maintains candidate awareness across all FFT bins and that low-confidence candidates trigger escalation. |
| DMA/shared-memory path | Pending implementation | Confirm the selected data mover, interconnect, memory map, cache-coherency method and error recovery. |
| Power measurement | Pending calibration | Confirm accessible monitor rails and update rate; use external calibrated equipment if necessary. |

| Item | Status | Required evidence/action |
|---|---|---|
| Synthesis, timing and resources | Not yet measured | Replace planning placeholders with actual synthesis/implementation reports. |
| Pd/Pfa, latency, throughput and energy | Not yet measured | Run the predefined test matrix and report measured results with test conditions and uncertainty. |
| Prior-art comparison and citations | Pending final source audit | Verify each reference and compare against the closest adaptive spectrum-sensing implementations. |
| Portal fields, team details and shared links | Team action required | Complete only with registered, accurate details; test all links and access permissions. |

Submission gate: do not call a technical item verified until its evidence exists. Do not modify a previously submitted contest package unless the organisers permit updates. Do not represent this proposal as guaranteeing kit selection, semifinalist status, energy savings, or a contest win.

[14] Microchip Technology Inc., “PolarFire SoC Icicle Kit Brochure,” DS00003611A, Aug. 2020 (device, memory, network, serial and expansion summary).

[15] PACTRON / Microchip, “ICICLE Kit Rev. 1.0 Schematics,” DVP-102-000536-001, dated June 23, 2020 (board connectivity, banks, clocks, power and peripheral wiring).

[16] Microchip Technology Inc., “PolarFire SoC FPGA Board Design Guidelines,” DS60001681H, 2026 edition (power, I/O, clocks, reset, DDR and signal-integrity constraints).